Super Store Sales Analysis Dashboard

Power BI Capstone Project

SUBMITTED BY

Rohan Thanvi

Roll Number: 23053625

Program: Power BI

1. Problem Statement

Retail superstores generate vast transactional data across product categories, customer segments, geographic regions, and shipping modes. Without a centralized analytical platform, decision-makers struggle to identify underperforming segments, understand regional sales distribution, and monitor profit trends. This project builds an interactive Power BI dashboard transforming raw Super Store data into actionable business intelligence.

2. Objective

To analyze Super Store sales data across regions, segments, and shipping modes — identifying revenue trends, profitability patterns, and operational insights — using Microsoft Power BI as the primary business intelligence tool.

- Visualize total sales, quantity, and profit through KPI cards for instant business overview.
- Analyze sales distribution across Consumer, Corporate, and Home Office customer segments.
- Track monthly profit and sales trends across 2019 and 2020 using comparative line charts.
- Map geographic sales distribution across global regions using integrated Bing Maps.
- Cover all 10 required visual types across two interactive dashboard pages.
- Enable dynamic regional filtering using a Region slicer across all dashboard visuals.

3. Solution & Features

A two-page interactive Power BI dashboard was built covering all 10 required visual types. Page 1 is the main sales overview; Page 2 provides regional and profitability analysis.

Visual	Type	Key Insight
Sales / Quantity / Profit	KPI Cards (3)	1.57M sales, 22K qty, 175.26K profit
Sum of Sales by Segment	Donut Chart	Consumer leads at 48% of total sales
Sum of Sales by Payment Mode	Donut Chart	COD dominates at 43% of all payments
Sum of Profit by Month & Year	Line Chart	Profit grew consistently 2020 vs 2019
Sum of Sales by Month & Year	Line Chart	Sales peaked in Nov-Dec both years
Geographic Sales Distribution	Map	Sales in North America & Europe
Sales by Sub-Category / Ship Mode / Category	Bar Charts (3)	Phones 0.20M; Std Class 0.33M; Off.Sup 0.64M
Sales Matrix — Region vs Category	Matrix	West region highest at 5.22M total
Monthly Profit Variance	Waterfall	Furniture shows consistent losses
Sales & Profit by Sub-Category	Scatter Plot	Technology — high sales, high profit
Detailed Sales Table	Table	Full category & sub-category breakdown
Region Filter	Slicer	Dynamic filtering across all visuals

4. Dashboard Screenshots

Page 1 presents the main sales overview with KPI cards, segment and payment donut charts, geographic map, monthly trend line charts, and category bar charts. Page 2 provides regional drill-down with a region slicer, sales matrix, waterfall, scatter plot, and a conditional-formatted data table.

Figure 1: Page 1 — Main Sales Overview (KPI Cards, Donut Charts, Line Charts, Map, Bar Charts)

5. Tech Stack

Technology	Purpose
Microsoft Power BI Desktop (2025)	Dashboard creation, data modeling, all visualizations
Microsoft Excel / CSV	Source data — Super Store Sales dataset from Kaggle
Power Query (M Language)	Data transformation and type-fixing inside Power BI
DAX (Data Analysis Expressions)	KPI measures, calculated fields and aggregations
Python 3.x (pandas)	Data cleaning and preprocessing before Power BI import
Bing Maps (integrated)	Geographic visualization of global sales distribution
GitHub (Public Repository)	Version control and project submission hosting

Figure 2: Page 2 — Regional & Profitability Analysis (Region Slicer, Matrix, Waterfall, Scatter Plot, Table)

6. Unique Points

- Two-page dashboard — Page 1 main overview, Page 2 regional profitability drill-down, mirroring real enterprise BI report structures.
- Full coverage of all 10 required visual types: KPI Cards, Line, Bar, Donut, Table, Matrix, Scatter, Map, Waterfall, and Slicer.
- Dual year comparison (2019 vs 2020) on line charts enabling direct Year-over-Year performance analysis.
- Conditional formatting on profit table — red highlighting for loss-making transactions provides instant visual alerts to decision-makers.
- Sales Matrix (Region vs Category) reveals West region leads at 5.22M; end-to-end pipeline from Python cleaning to interactive dashboard.

7. Key Insights & Findings

- Office Supplies is the top revenue category at 0.64M, followed by Technology (0.47M) and Furniture (0.45M).
- Consumer segment dominates at 48% of total sales, making it the primary target customer group.
- Standard Class shipping accounts for 3.45K orders — the most preferred ship mode by far.
- Phones are the top sub-category by revenue at 0.20M, followed by Chairs (0.18M) and Binders (0.17M).
- Profit grew consistently through 2020 vs 2019, with the highest spike observed in Q4 (Oct-Dec).
- West region leads all regions at 5.22M in total sales across all categories per the sales matrix.

8. Future Improvements

- Connect to a live SQL database or cloud data warehouse for real-time dashboard refresh.
- Add DAX-based profit margin % and Return on Sales (ROS) measures per category and region.
- Build a Customer Segmentation page using RFM (Recency, Frequency, Monetary) analysis.
- Publish to Power BI Service with role-based access so regional managers see only their data.
- Add forecasting visuals using Power BI's Analytics pane to predict next quarter revenue.
- Add a Returns & Refunds Analysis page to identify and reduce loss-causing transactions.

9. Conclusion

This Power BI Capstone Project successfully demonstrates the ability to transform raw Super Store sales data into a meaningful, interactive two-page business intelligence dashboard. Through Python-based data preprocessing, Power Query transformation, relational data modeling, DAX measures, and professional dashboard design covering all 10 required visual types — including dual-year trend comparison, conditional profit formatting, and geographic sales mapping — this project reflects a comprehensive, industry-ready understanding of the complete Power BI workflow, from raw data to actionable business insights.